

First Experimental Demonstration of Dual-sided N/P FETs in Flip FET (FFET) on 300 mm Wafers

Stacked Transistor Technology in Sub-1 nm Nodes

Presented for Advanced CMOS Devices and Technology

Mirudhula J

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Outline

Introduction

FFET Concept

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Scaling Challenge

- ▶ Moore's Law: transistor count doubles every ~ 2 years
- ▶ Conventional planar scaling is essentially exhausted
- ▶ FinFETs replaced planar MOSFETs; Gate-All-Around (GAA) nanosheets are replacing FinFETs at 3 nm and below
- ▶ **Next Problem:** even GAA nanosheets run out of lateral area
- ▶ **Solution:** stack transistors *vertically* — N-type FET on one side, P-type FET on the other side of the same wafer

Core idea of this paper

Exploit **both sides** of a silicon wafer to double transistor density without shrinking the gate length further

The 3D Transistor Landscape

Existing 3D stacking approaches:

- ▶ **Monolithic CFET** — N and P grown sequentially on one face; needs high aspect-ratio vertical patterning; tight thermal budget
- ▶ **Sequential CFET** — N and P on separate wafers, then bonded; avoids thermal issues but devices are asymmetric
- ▶ **FFET (this work)** — flip the wafer, build second device type on the back; no high aspect-ratio patterning needed

Key FFET advantages

- ▶ Standard aspect ratio (same as FinFET)
- ▶ N/P processes fully decoupled
- ▶ Natural split-gate capability
- ▶ Easy multi- V_t tuning
- ▶ Symmetric devices and interconnects on both sides

What Exactly is a Flip FET?

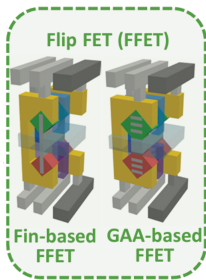


Figure 1: FFET

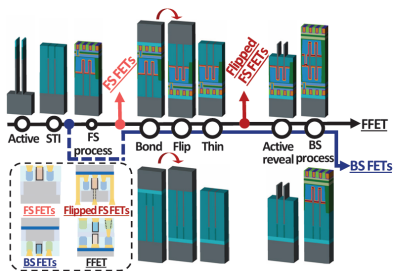


Figure 2: process flow

- ▶ FFET stacks the **frontside NFET** and **backside PFET** back-to-back on the same wafer
- ▶ After frontside processing, the wafer is bonded, flipped, thinned, and processed from the backside

Three Critical New Process Modules

► Bonding & Thinning

- Wafer bonding + substrate removal

► BS Fin Optimization

- Straight backside fins improve device control

► Overlay Correction

- Alignment error reduced to 4.4 nm

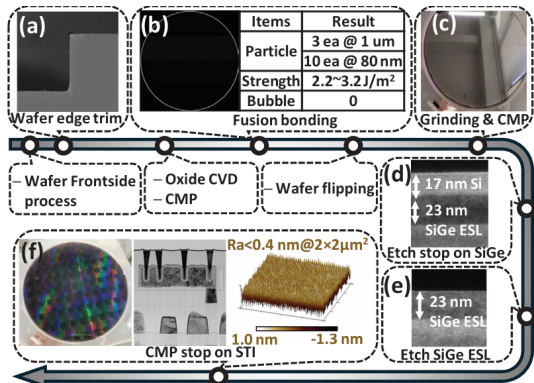
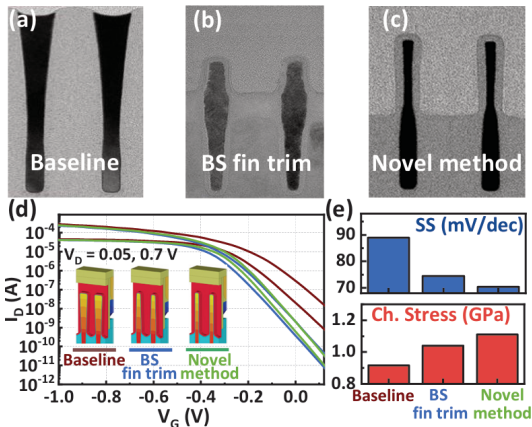


Figure: Key FFET fabrication modules

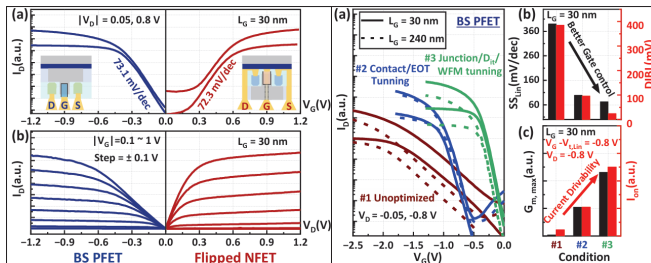
Bonding, thinning, backside fin optimization, and overlay correction are the major process innovations enabling FFET fabrication.

BS Fin Profile Optimization



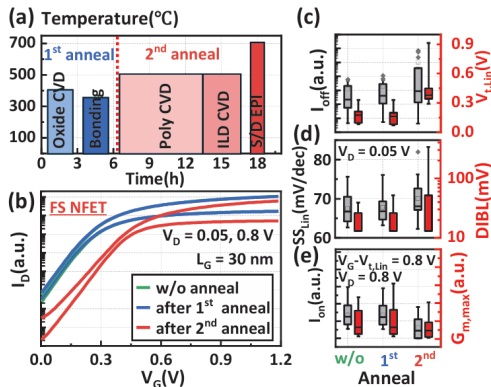
- ▶ Reverse-tapered fin $\rightarrow$ weak gate control, worse SS
- ▶ IBE improves fin shape but damages crystal
- ▶ **Novel method: straight fin + intact crystal**
- ▶ Best SS and highest channel stress $\rightarrow$ improved PFET mobility

Key Electrical Results



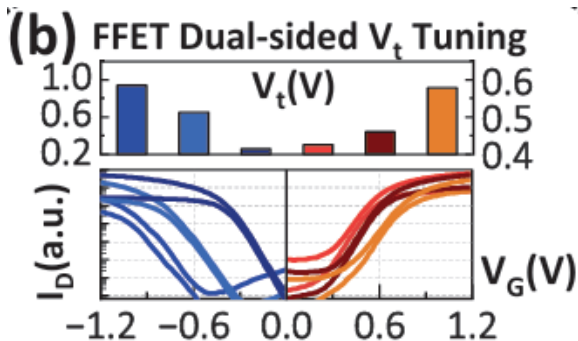
- ▶ BS PFET and flipped FS NFET show nearly matched performance
- ▶ Final optimized BS PFET: $SS = 73.1 \text{ mV/dec}$, $DIBL = 24 \text{ mV}$
- ▶ Optimization: contact/EOT tuning + WFM/D_{it}/junction engineering
- ▶ Confirms FFET CMOS viability with symmetric device behavior

Thermal Budget Impact on FS NFET



- ▶ Additional BS thermal steps affect already-fabricated FS NFETs
- ▶ 1st anneal: minor degradation
- ▶ 2nd anneal: noticeable V_t shift and SS degradation
- ▶ Motivates FFET multi-flipping strategy

Advantage 1 — Multi- V_t Tunability



- ▶ FFET supports independent frontside/backside WFM tuning
- ▶ V_t tuning range of nearly 500 mV demonstrated
- ▶ Achieved using standard industry process flow
- ▶ Avoids complex vertical patterning required in CFET

Advantage 2 — Split-Gate and Dual-sided CMOS

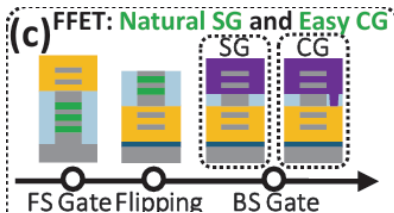


Figure: Split-Gate

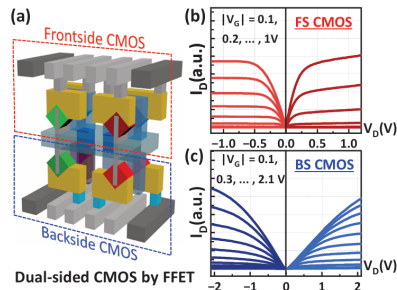


Figure: Dual-Sided CMOS

- ▶ FFET naturally supports independent split-gate operation
- ▶ Common-gate achieved by simple gate merge process
- ▶ Both FS and BS can independently host N/P devices
- ▶ Enables flexible true 3D CMOS integration

FFET vs. CFET: Side-by-Side Comparison

Feature	Mono. CFET	Seq. CFET	FFET (this work)
Self-aligned active	✓	×	✓
Process aspect ratio	High	Low	Low
Thermal budget risk	Serious	Serious	Manageable*
Multi- V_t	Difficult	Industry std.	Industry std.
Split gate	Complex	Complex	Natural
Interconnect symmetry	Asymmetric	Asymmetric	Symmetric
Device symmetry	Asymmetric	Asymmetric	Symmetric

* Solvable by multi-flipping strategy

Summary

FFET wins on every dimension except the thermal budget issue, which has a proposed architectural solution (multi-flipping)

Conclusions

- ▶ First demonstration of dual-sided FFET CMOS on 300 mm wafers
- ▶ BS PFET and flipped FS NFET achieved matched performance ($SS \sim 72\text{--}73 \text{ mV/dec}$, $\text{DIBL} = 24 \text{ mV}$)
- ▶ Key FFET modules validated:
 - ▷ Backside device integration
 - ▷ Dual-sided V_t tuning
 - ▷ Natural split-gate and dual-sided CMOS
- ▶ Main remaining challenge:
 - ▷ Thermal budget management
 - ▷ Further scaling and BEOL integration

Takeaway

FFET offers a scalable and process-friendly alternative to CFET, with strong advantages in symmetry, flexibility, and 3D integration.

Thank You

Questions?